

MANAGEMENT OF SALICYLATE POISONING

(REFER TO ANNEX A FOR WORKFLOW)

PEARLS

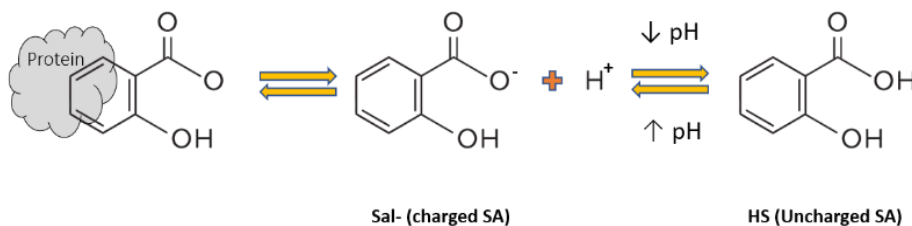
1. Different formulations have different concentrations of Salicylic Acid (SA) (**ANNEX B**)
2. Goal = enhance elimination of salicylate + reverse tissue salicylate burden
3. Avoid intubation if possible; If patient requires intubation, consider IV NaHCO_3 2 mmol/kg bolus before intubation to buffer serum pH
4. POCT (urine dipstick + ABG/ME) can help determine severity of salicylate poisoning while serum salicylic acid levels are being processed
5. Do not confuse SI units (10 mg/L = 1 mg/dL); **NUH = mg/L**
6. Alkalinization of urine will not occur in the presence of hypokalemia (potassium must be replaced first)

PHARMACOKINETICS AND TOXICOKINETICS

- SA is rapidly absorbed by the GI tract after ingestion. Rate of absorption is affected by:
 1. Dosage formulation
 2. Gastric/intestinal pH
 3. Gastric emptying time
- Duration to peak plasma levels is longer with higher doses of SA due to pylorospasm and bezoar formation with resultant delayed gastric emptying and systemic absorption

	TIME TO PEAK PLASMA LEVEL	
	Standard formulation	Enteric Coated (EC) formulation
Therapeutic dose	15 min – 2 hours	8 – 14 hours
Toxic dose	4 – 6 hours	< 72 hours

- Peak plasma levels typically occur 4 – 6 hours after a single ingestion of a toxic dose of standard formulation



- Tissue levels of salicylate (not plasma levels) determine severity of salicylism
- About 80 – 95% of SA is bound to albumin; the remaining free SA exists in a charged [**Sal⁻**] and an uncharged form [**HS**] → it is this form that readily crosses into tissues, worsening salicylism
- Metabolism of SA is non-linear → enzymes are saturated and higher doses have a longer $T_{1/2}$ (from 3h to 20h)
- Patients with chronic SA exposure (> 100 mg/kg/day ASA for ≥ 2 days) are at risk of poisoning at lower doses:
 - Tissue burden is elevated at baseline
 - Albumin and metabolic enzymes already saturated

PATHOPHYSIOLOGY OF SALICYLATE POISONING

CNS

1. Stimulation of chemoreceptors in medulla → nausea & vomiting
2. Direct stimulation of respiratory centre in medulla → hyperventilation, ↑ insensible losses
 - Extremely high CNS levels of SA depresses respiratory centre → hypoventilation

Metabolic

1. Interferes with aerobic metabolism:
 - Shift to glycolysis for energy production (ineffective process)
 - ↑ heat production → hyperpyrexia, diaphoresis & dehydration
 - ↑ O₂ consumption & ↑ metabolic production of CO₂
2. ↓ glycogen stores → hypoglycemia & neuroglycopenia (can occur with normal peripheral glucose)
 - Accumulation of lactate, ketones, and other organic acids → 1° metabolic acidosis
3. 1° respiratory alkalosis
 - ↑ renal losses of HCO₃ (alkaluria) (with conservation of Cl⁻) → NAGMA → contributes to ↑ acidosis
 - With systemic alkalosis → renal conservation of H⁺ over K⁺ → hypokalemia
 - Worsening hypokalemia over time → renal conservation of K⁺ over H⁺ → paradoxical aciduria

Direct Toxicity

1. Haem – Leukocytosis, inhibition of platelet function, disturbance of clotting factors → coagulopathy
2. CVS – Non-cardiogenic pulmonary edema
3. MSK – Muscle paratonia → rigidity → rhabdomyolysis
4. Renal – Acute renal injury
5. CNS – Cerebral edema, seizures

INVESTIGATIONS

- ABG/ME, H/C, UC10 (for initial evaluation if dose > 125 mg/kg)
- FBC, PT/INR/PTT
- RP2, Mg
- LFT
- Lactate
- Salicylic acid (if ≥ 4 hours from ingestion)
- ECG
- Consider the following (if appropriate):
 - Acetaminophen, ethanol (if history of co-ingestion)
 - Urine drug screen (if degree of AMS unexplained by history provided)

Conversion (SI units)

NUH lab unit = **mg/L**

▫ 1 mg/L = 0.1 mg/dL

▫ 1 mg/L = 0.0072 mmol/L

▫ 1 mmol/L = 138.12 mg/L

MANAGEMENT OF SALICYLATE POISONING

1. Emergency stabilization
2. Decontamination
3. Enhanced elimination
4. Supportive therapy

EMERGENCY STABILIZATION

Airway Breathing	▫ Direct CNS insult to respiratory centre, non-cardiogenic pulmonary edema ▫ Avoid mechanical ventilation → respiratory acidosis from apnea during RSI results in mixed acidosis → favours uncharged form of free SA → tissue toxicity ▫ In event of cardiorespiratory collapse, intubate with considerations: 1. IV NaHCO₃ <u>2mmol/kg</u> to optimize plasma pH before intubation 2. Short acting paralysis (eg. Succinylcholine) 3. Most experienced operator for intubation 4. Employ use of video laryngoscope if available ▫ Post-intubation 1. Set ventilation rate to pre-intubation rate 2. Volume-control for guaranteed minute ventilation
Seizures	▫ Causes: 1. Neuroglycopenia (with normal peripheral glucose) 2. Hypocalcemia (lowers seizure threshold) 3. Cerebral edema ▫ Treatment: 1. IV lorazepam <u>0.1mg/kg</u> (max. 4mg) slow bolus 2. IV D10% <u>5ml/kg</u> bolus 3. IV 10% Ca gluconate (0.23mmol/ml) solution <u>0.5ml/kg</u> (max. 20ml in 100ml 0.9% NaCl) over 20 mins

DECONTAMINATION

Gastric decontamination	▫ Activated charcoal <u>1 g/kg</u> (max 50g) 1. Up to 8h after ingestion of solid formulation 2. Up to 12h after ingestion of EC/SR formulation
Excluding bezoar	▫ Prolongs GI absorption, delay in peak serum levels, prolongs toxicity ▫ Consider evaluating for bezoars if plasma levels rise despite repeated doses of activated charcoal ▫ Diagnosed on: 1. Gastroscopy 2. Ultrasound 3. CT abdomen

ENHANCED ELIMINATION

Multiple doses of Activated Charcoal	▫ Repeated doses of Activated Charcoal <u>0.5 – 1 g/kg</u> every 2 – 4h till symptoms improve or plasma levels down-trending ▫ Consider for: 1. SR/EC formulation 2. Suspected pharmacobezoar formation ▫ Contraindications: 1. Ileus or intestinal obstruction 2. Depressed mental state with no airway protection
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Urinary alkalinization *alkalinization of urine will not occur with hypokalemia (it must be corrected first)	▫ Indicated for moderate to severe toxicity (paradoxical aciduria, urine pH < 6) ▫ Anticipate hypokalemia with administration of IV NaHCO₃ ▫ Loading dose: IV 4.2% NaHCO₃ <u>1 mmol/kg</u> over 30 mins ▫ Maintenance: 1. <u>150ml of 8.4% NaHCO₃ + 20mmol of KCl + D5% solution</u> (final volume of 1L) ▪ Alternative: D5% + 0.9% NaCl (with KCl) 2. Titrate infusion to keep urine pH > 7.5 (start @ 1.5X maintenance rate) ▫ Monitoring: 2-hourly ABG/ME & hourly UC10 1. Aim pH 7.5 – 7.55 2. Aim urine pH > 7.5 3. Aim serum potassium 4.0 – 4.5 mmol/L
Hemodialysis	▫ Indicated for: 1. Plasma levels > 700 mg/L 2. Severe signs and symptoms 3. Altered mental status 4. Hypoxemia requiring oxygen 5. Impaired renal function (Creatinine 2x upper limit for age / sex) 6. After intervention, ▪ Persistent salicylate levels > 700 mg/L ▪ Systemic pH < 7.2 ▫ Termination of dialysis when plasma levels < 190 mg/L ▫ Watch for rebound toxicity after termination of dialysis

SUPPORTIVE THERAPY

Metabolic acidosis	▫ Aim systemic pH 7.5 – 7.55 1. If systemic pH < 7.3 → IV 4.2% NaHCO₃ <u>1mmol/kg</u> slow bolus 2. If systemic pH < 7.2 → consider hemodialysis
Hypoglycemia	▫ Secondary to depletion of glycogen stores after activation of glycolysis ▫ Neuroglycopenia can occur in the presence of normal plasma glucose ▫ IV rescue: 1. Asymptomatic/symptomatic (mild) hypoglycemia: IV D10% <u>2ml/kg</u> 2. Neuroglycopenia with seizures: IV D10% <u>5ml/kg</u>
Hypokalemia	▫ Expected in salicylate poisoning, aggravated by NaHCO₃ infusion ▫ Be cautious with replacement in oliguric/anuric patients
Dehydration	▫ Hydrate with 1.5X maintenance fluid volume (As per urinary alkalinization fluid)
Coagulopathy	▫ Hypoprothrombinemia, inhibition of platelet aggregation and thrombocytopenia ▫ Administration of IV Vitamin K <u>0.3mg/kg</u> (max 10mg)

OTHERS

- All cases of drug overdose should be reported to the police unless drug overdose was an attempt at suicide
- See **ANNEX C**

SEVERITY OF SALICYLATE TOXICITY

- Assessment of the severity of salicylate poisoning should be guided by:
 1. Plasma salicylate levels (≥ 4 hours from ingestion)
 - Levels should be done every 2 hours until peak plasma levels obtained
 - Salicylate levels might not peak until > 12 hours after overdose on EC formulations
 - Levels taken 6 hours from acute ingestion roughly correlates with severity of toxicity
 2. Acid-base status (see table below)
 - ABG/ME and UC10 should be performed in symptomatic patients (while awaiting salicylate levels) as objective assessment of salicylate toxicity
 3. Duration of poisoning
 4. Signs and symptoms of poisoning (see table below)

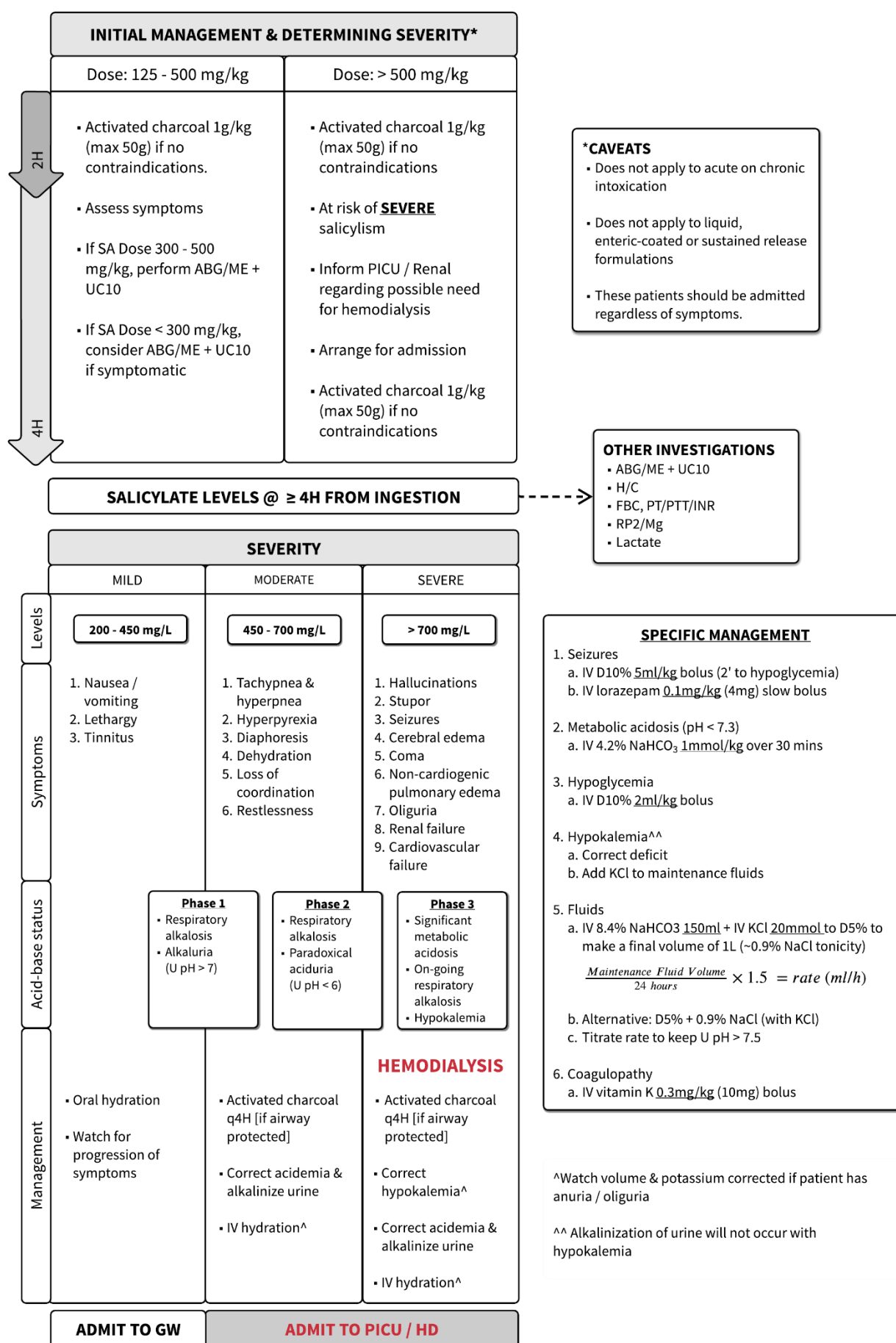
SEVERITY	TIME FROM INGESTION	EST. PLASMA LEVELS	SYMPTOMS	PHASE	POCT CHANGES
Mild	1 – 2h	200 – 450 mg/L	Nausea & vomiting Lethargy Tinnitus*		
Moderate	6 – 18h	450 – 700 mg/L	Tachypnea & Hyperpnea Hyperpyrexia Diaphoresis Dehydration Loss of coordination Restlessness	1	Respiratory alkalosis Alkaluria
				2	Respiratory alkalosis Paradoxical aciduria
Severe	12 – 24h	> 700 mg/L	Hallucinations Stupor Seizures Cerebral edema Coma Non-cardiogenic pulmonary edema Oliguria Renal failure Cardiovascular failure	3	Metabolic acidosis Respiratory alkalosis Hypokalemia

*Tinnitus can be observed at high normal therapeutic drug levels

DISPOSITION

- Type of ward dependent on severity of salicylic toxicity
 - Mild: General Ward
 - Moderate: HD
 - Severe: PICU
- Discharge if PEAK serum levels < 200 mg/L

ANNEX A. FLOWCHART FOR MANAGEMENT OF SALICYLATE POISONING



ANNEX B. DIFFERENT FORMULATIONS OF SALICYLIC ACID

SALICYLIC ACID FORM	REMARKS	ACF	INTERVENTION LEVEL
Aspirin "Acetyl Salicylic Acid (ASA)"	<u>Chronic exposure</u> = > 100 mg/kg/day for ≥ 2 days		150 mg/kg (6.5 g)
Salicylic Acid	Topical keratolytic & wart remover	1.3043	115 mg/kg (5 g)
Oil of Wintergreen "Methyl Salicylate"	Rapid severe SA toxicity Extremely concentrated (for liquid) Highly lipophilic	1.4017	<u>Methyl Salicylate</u> 105 mg/kg (4.5 g) <u>Oil of Wintergreen</u> < 6y: a lick > 6y: > 4ml
Bismuth subsalicylate	Anti-diarrheal	0.4975	300 mg/kg (13 g)
Magnesium subsalicylate (+ caffeine)	<u>Chronic exposure</u> = > 80 mg/kg/day anhydrous magnesium salicylate for ≥ 2 days Anti-bloat	1.2068	100 mg/kg (5.3 g)

ASPIRIN CONVERSION FACTOR (ACF)

Solid Compounds

$$\frac{\text{Molecular weight of aspirin}}{\text{Molecular weight of compound}} = ACF$$

$$\text{Non - aspirin salicylate (mg)} \times ACF = \text{Equivalent aspirin amount (mg)}$$

Liquid Compounds

$$\frac{\text{Molecular weight of aspirin}}{\text{Molecular weight of compound}} \times \text{density of liquid} = ACF$$

$$\text{Non - aspirin salicylate (ml)} \times ACF = \text{Equivalent aspirin amount (g)}$$

**ANNEX C. DECRIMINALISATION OF ATTEMPTED SUICIDE AND NEW WORKFLOW
FOR CASE MANAGEMENT AND DATA TRACKING “MOH CIRCULAR
57/2019”**

- Penal code amended in May 2019 – suicidal persons in distress should receive help and not be penalised by the criminal justice system
- SOP:
 - Patients will be sent to the hospital if SCDF or SPF assesses that the patient requires medical attention resulting from physical injuries or overdose of medication
 - ED nurse will sign bottom right-hand corner of 1st page of NP306; endorsed carbon copy will be handed back to police.
 - Patient indicated as “Sec 7 of MHCTA (Attempted Suicide)” in the NP306

REFERENCES:

1. Dargan, P I (2002). *“An evidence based flowchart to guide the management of acute salicylate (aspirin) overdose.”* Emergency Medicine Journal, 19(3), 206–209. doi:10.1136/emj.19.3.206
2. Ingelfinger, Julie R.; Palmer, Biff F.; Clegg, Deborah J. (2020). *“Salicylate Toxicity”*. New England Journal of Medicine, 382(26), 2544–2555. doi:10.1056/NEJMra2010852
3. *“Salicylate”*, www.toxinz.com (November 2020)